

Summarize

FM6373C is a full-color LED display driver chip designed for high-refresh applications. The FM6373C has a built-in 16-bit shift register that converts serial input data into grayscale pixels for each output channel. FM6373C 16 constant current output channels output current value is not affected by the output terminal load voltage, provide consistent constant output current, not affected by PCB board, has excellent anti-interference. FM6373C can not only adjust the current of each output stage of FM6373C by external resistors of different resistance values, but also adjust the channel current gain by software to control the brightness of LED display. FM6373C adopts rich EPWM patent display technology, the display effect is better and more delicate.

FM6373C supports up to 64 line scanning LED screen; through technological innovation, it can effectively solve the problems of low gray block, color cast, pitting, first line dark, high contrast coupling, cross-border chromatic aberration, etc.; internal integrated open circuit detection function; can effectively solve the cross problem caused by LED bad dots. In addition, the FM6373C's built-in blanking/clamping circuit, combined with different configuration bits, can greatly enhance the blanking/clamping capabilities, effectively solve the upper and lower ghost and text ghost phenomena, and also greatly improve cross-version display defects such as coupling, high contrast, and medium contrast coupling; the FM6373C can internally generate GCLK signals to optimize system performance; it also has excellent dynamic adaptation and channel adaptation, resulting in better accuracy.

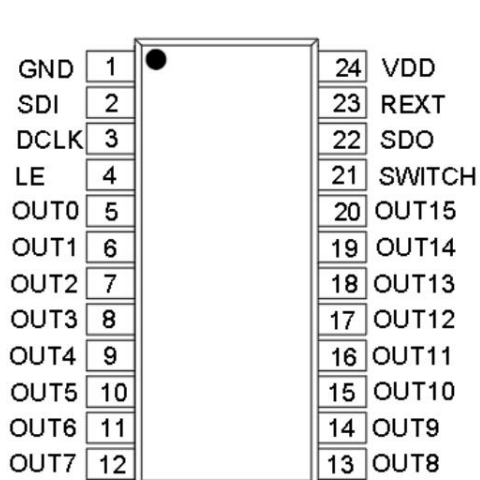
Application area

- LED display with high brush color screen

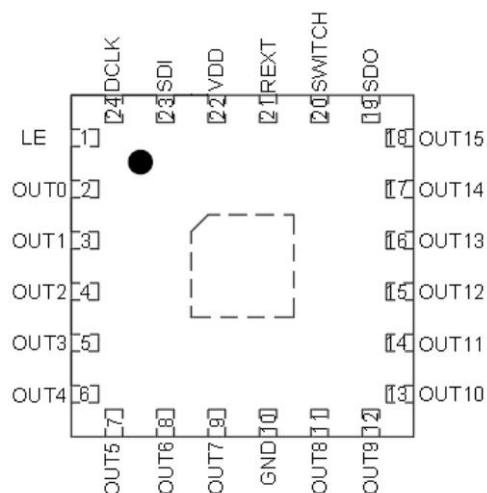
Characteristic

- Constant current output range:
- Operating voltage at 5 volts: 0.5~25mA
- Operating voltage at 4.2 volts: 0.5~18mA
- Extremely accurate current output values:
 - Maximum difference between channels: $\pm 1.75\%$
 - Maximum difference between chips: $\pm 1.75\%$
- Built-in 32K SRAM memory, can support up to 64 scan time-sharing multitask scan
- EPWM gray scale control technology improves low gray visual effect (FM exclusive patent)
- 8-bit current gain adjustment, 22%~200%
- Effectively improve cross-plate coupling, high contrast coupling problems
- Constant current inflection point 2 adjustable
- Built-in clamp, blanking enhancement
- LED bad lights excluded:
 - Eliminate the cross phenomenon caused by LED bad dots
 - LED open circuit detection
- Internally generated GCLK signal
- Frequency doubling function
 - Black screen energy saving, effectively reduce black screen power consumption
- DCLK clock input up to 15MHz

Package form

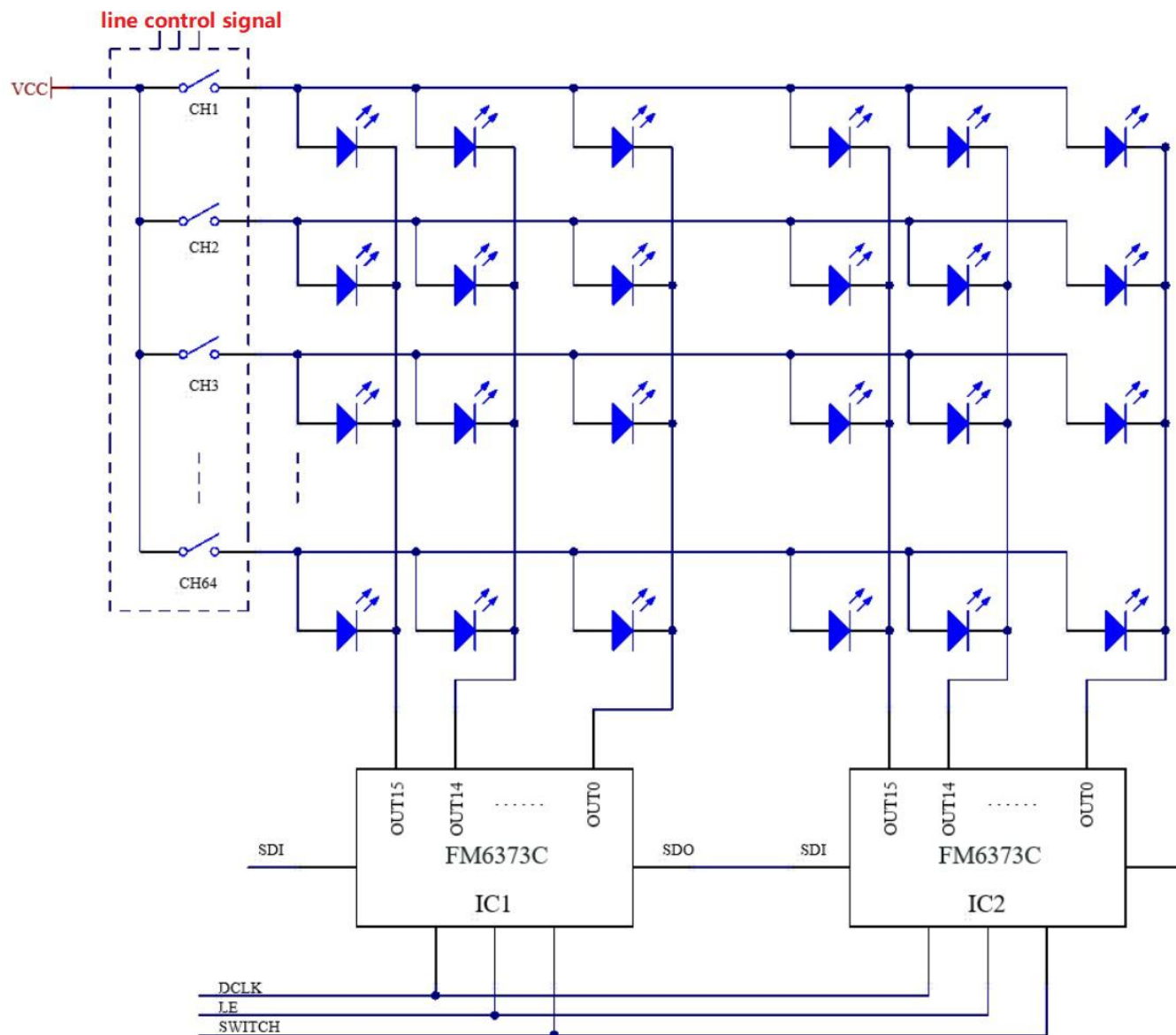


➤ SSOP-24(e=0.635mm)



BN: QFN-24-4×4-0.5mm

Typical application circuit of scanning screen

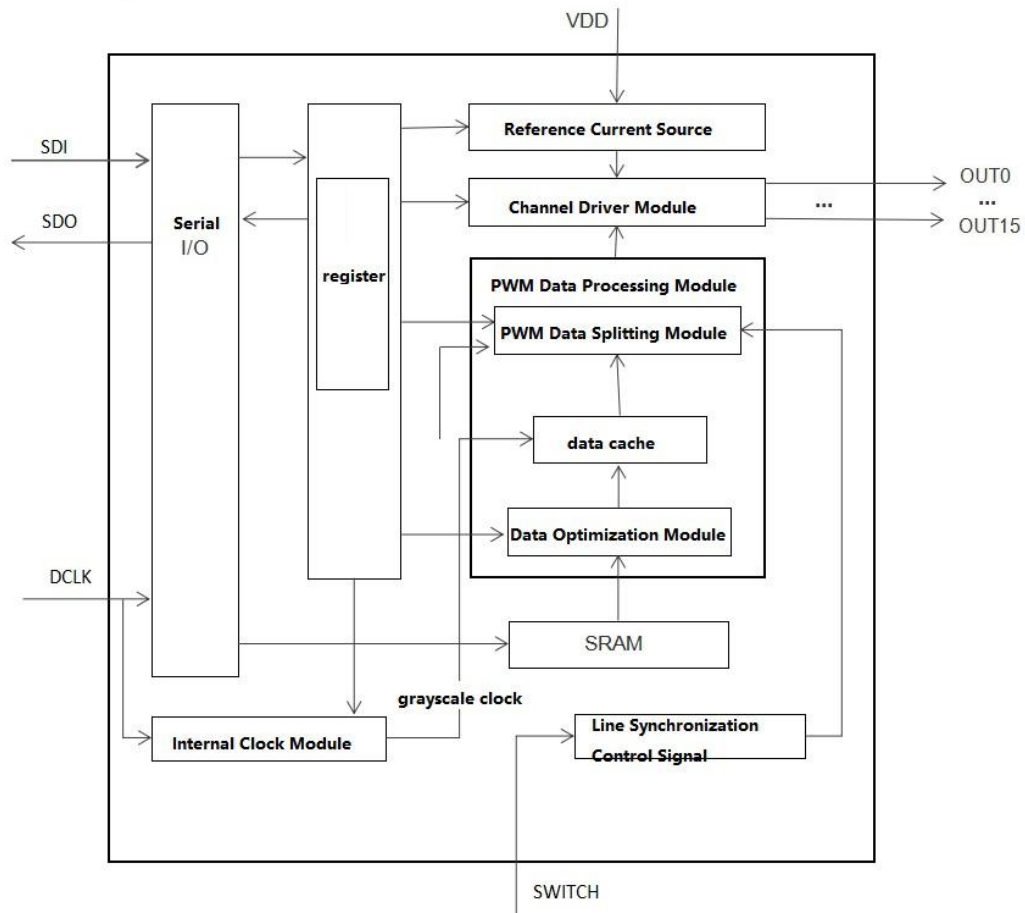


The above figure shows the proposed application architecture for a 64-line scan screen. A command consisting of LE and DCLK transmits gray scale data via the SDI and SDO pins. The 16 output channels (OUT0 to OUT15) deliver PWM data to each scan line at different points in time, so switches for time-division multitasking must be available for each scan line.

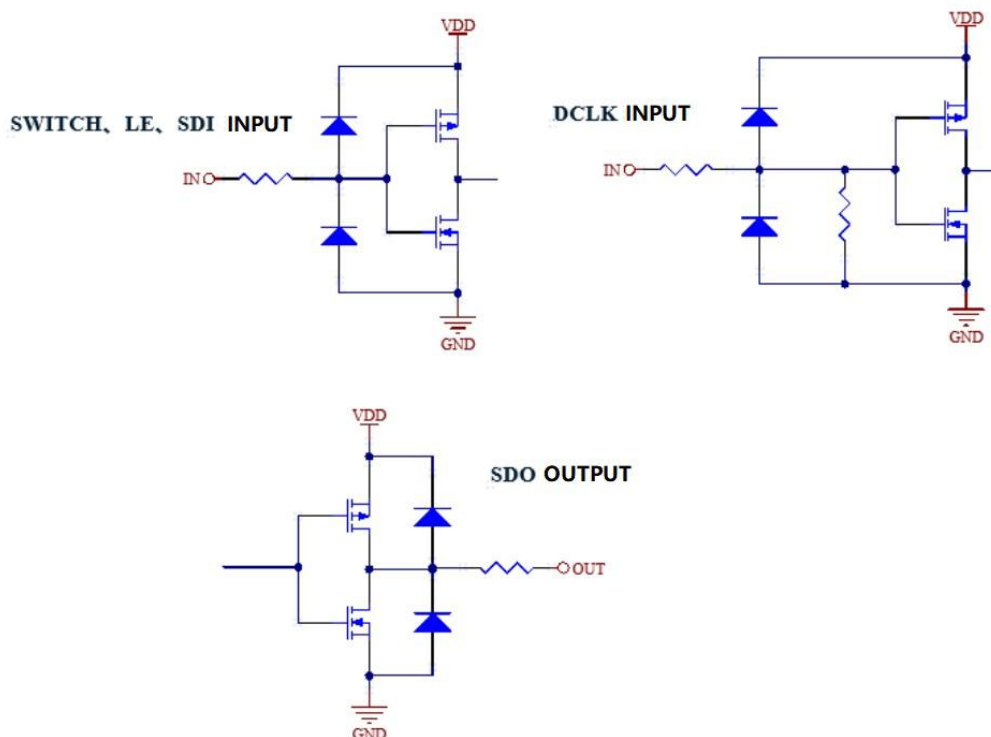
Pin description

pin name	function description
GND	ground terminal
SDI	serial data input
DCLK	data clock signal input terminal
LE	data latch input
OUT0~OUT15	Channel constant current output
SWITCH	newline signal
SDO	serial data output
REXT	External resistor input terminal; this resistor can set the output current of the channel
VDD	power supply terminal

Functional block diagram



Port equivalent circuit



Limit parameters

parameter	symbol	parameter scale	unit
mains input	V_{CC}	0~6	V
Input Voltage (SDI, CLK, LE, SWITCH)	V_{LED}	-0.4~ V_{DD} +0.4	V
output withstand voltage	V_{ds}	-0.4~ V_{DD} +0.4	V
output side current	I_{OMAX}	+25	mA
earth terminal current	D_{IO}	500	mA
junction temperature	$T_{j,max}$	150*	° C
Ambient temperature at which the chip operates	T_{opr}	-40~+85	° C
Ambient temperature at which chips are stored	T_{stg}	-55~+150	° C

* The closer to this maximum range, the shorter the life of the chip and the lower the reliability; when working beyond this maximum limit, it will affect the operation of the chip and cause damage, so the recommended joint temperature of the chip when working is within 125 ° C.

Electrical Parameter Characteristics($V_{DD} = 5.0V$, $T_a = 25^\circ C$)

character		symbol	measurement condition	least value	typical value	maximum	unit
output withstand voltage		Vds	OUT0 ~ OUT15	-	-	VDD +0.3	V
input level	high level	VIH	Ta=-40~85° C	$0.7 \times VDD$	-	VDD	V
	low level	VIL	Ta=-40~85° C	GND	-	$0.3 \times VDD$	V
output terminal voltage	SDO	VOH	IOH=-1.0mA	VDD-0.4	-	-	V
		VOL	IOL=+1.0mA	-	-	0.4	V
Current Offset (Channel to Channel)		dIOUT1		-	± 1.75	-	%
Current offset (chip-to-chip)		dIOUT2		-	± 1.75	-	%
Current offset vs. output voltage		%/Vds	V DS between 1.0V and 3.0V, Rext =1.3KΩ @5mA	-	± 0.1	-	%/V
Current offset vs. supply voltage		%/VDD	V DD between 4.5V and 5.5V, Rext =1.3KΩ @5mA	-	± 0.1	-	%/V
pull-down resistor		RIN (down)	LE		210		K Ω
voltage source current output	"Off" (SDI=DCLK=SWITCH CH =0Hz)	IDD1(off)	Rext =Open, OUT0~OUT15 =Off	-	2.0	-	mA

Test circuits for electrical parameters

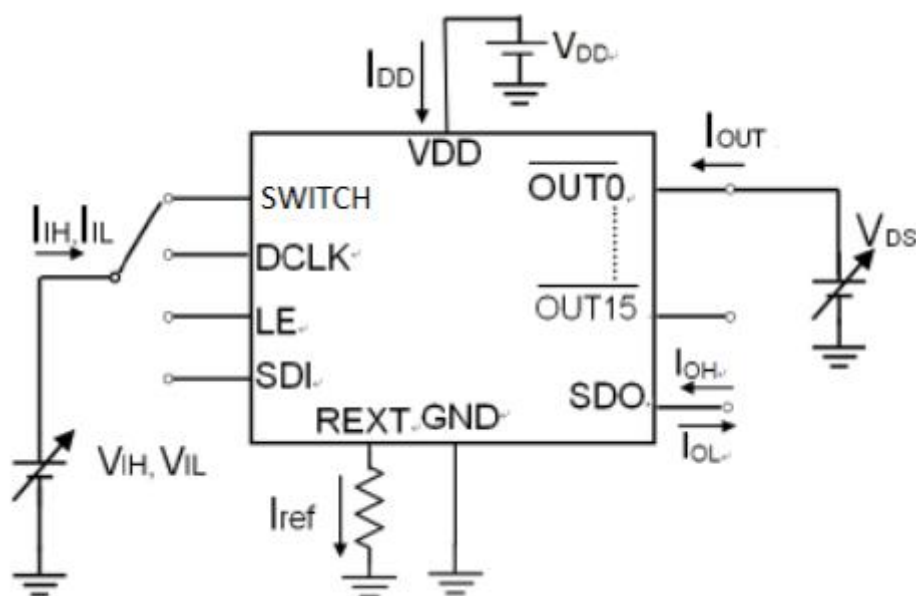


Fig.2

Dynamic electrical parameter characteristics (T = 25°C, V_{DD} = 5.0V)

character		symbol	measurement condition	least value	general value	maximum	unit
set time	SDI-DCLK ↑	tSU0**	VDD=5.0V VIH=VDD VIL=GND Rext=1.3K Ω Vds=1V RL=300 Ω CL=10pF C1=100nF C2=10 μ F CSDO=10pF VLED=4.0V	3	-	-	ns
	LE ↑ - DCLK ↑	tSU1**		7	-	-	ns
storage time	DCLK ↑ - SDI ↓	tH0**		4	-	-	ns
	DCLK ↑ - LE ↓	tH1**		7	-	-	ns
delay time	DCLK - SDO	tPD0**		-	25	-	ns
pulse width	LE	tw(LE)**		15	-	-	ns
current output rise time		tOR**		-	25	35	ns
current output fall time		tOF**		-	25	35	ns

** Refer to timing waveform diagram

Test Circuit for Dynamic Electrical Parameter Characterization

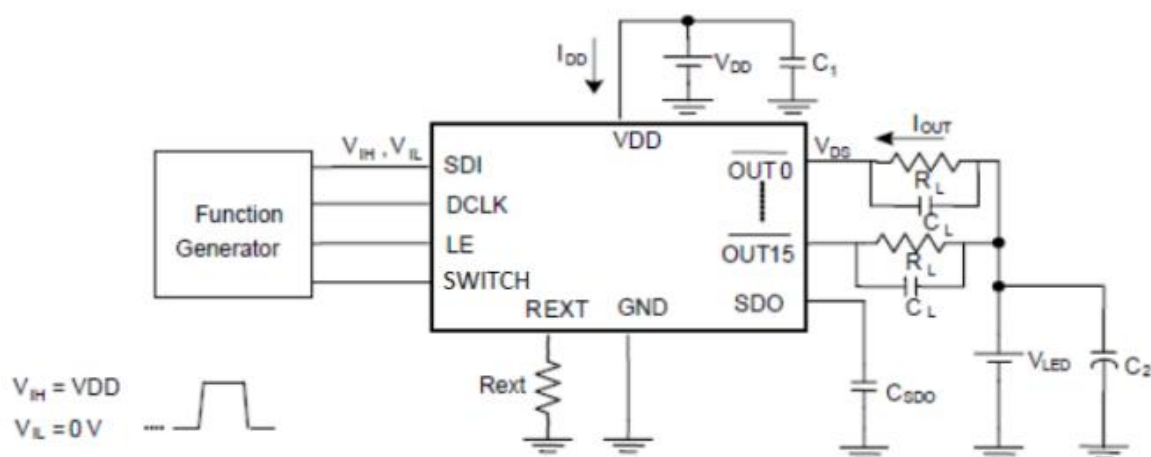
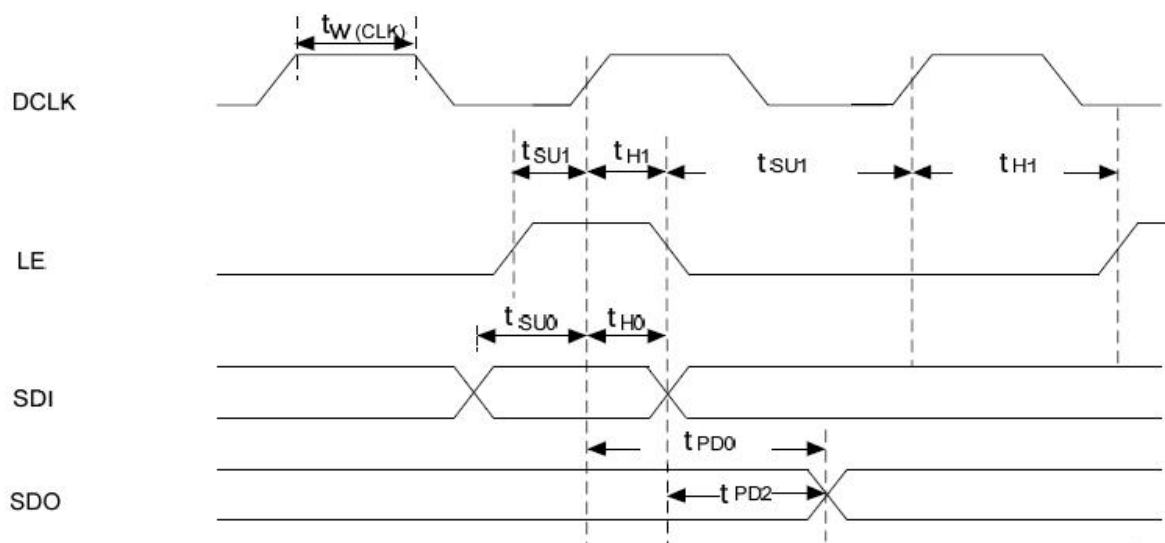
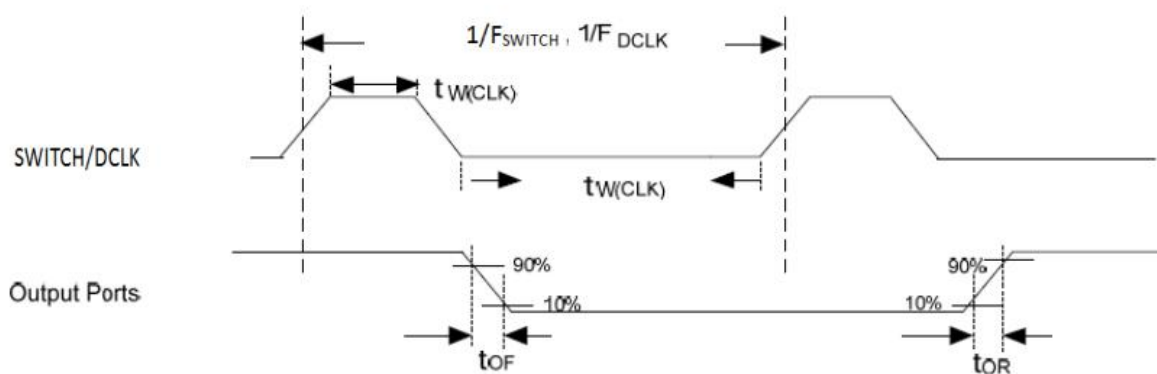


Fig. 3

Timing waveform diagram



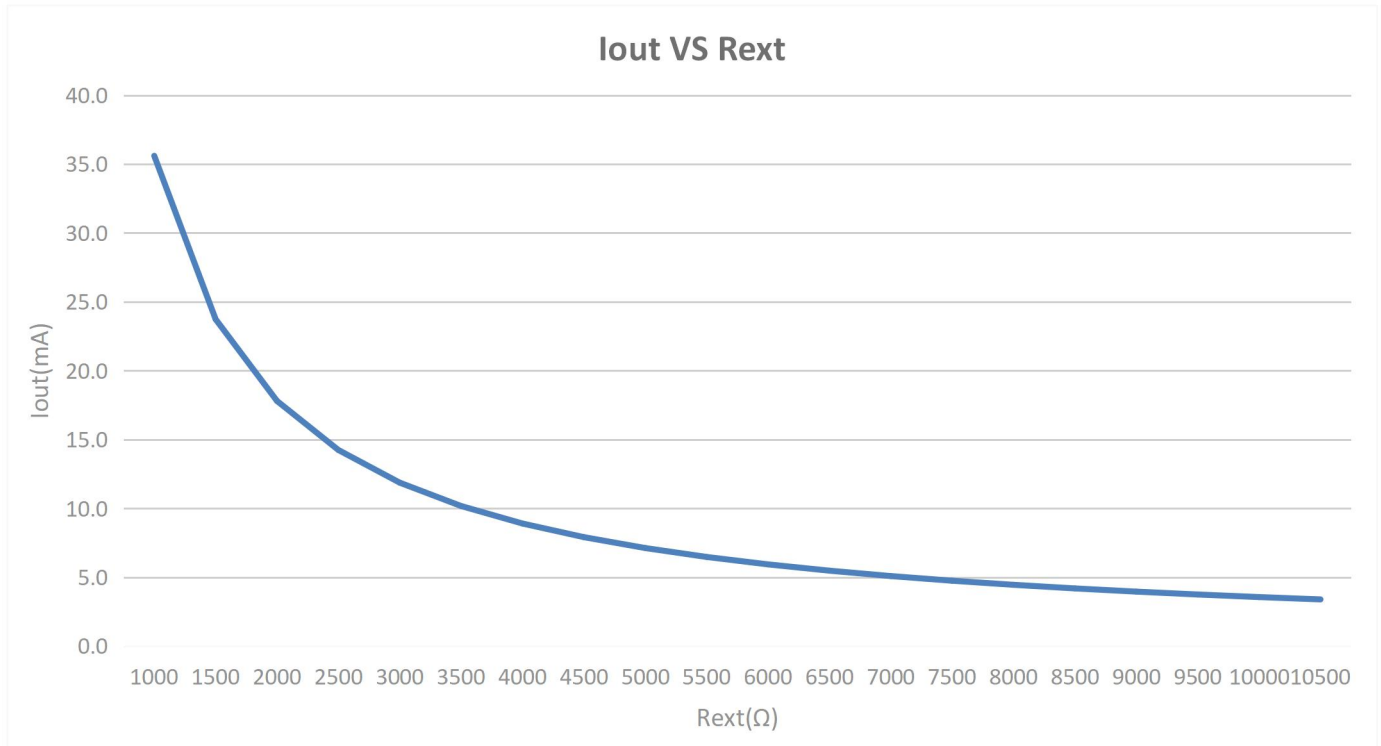
(1)



(2)

Application message

The relationship between output current and Rext resistance is shown below (reg 0x1C [7]=1, reg 0x1C [6:0]= 111111):



The FM6373C uses precise current control technology to achieve very small differences in current between channels and even between chips.

The maximum current difference between channels is less than $\pm 1.75\%$, while the maximum current difference between chips is less than $\pm 1.75\%$.

2. It has current output characteristics that are not affected by the channel terminal voltage. It is recommended that the lowest channel voltage (Vout_min) reference value for maintaining constant output current of the chip is as follows, where Vds is an internal parameter of the chip and can be configured through a register. The typical value is 0.36V, so the constant current inflection point of the chip is.

3, output channel current, by connecting an external resistor, with Register 1C[7:0] to adjust the voltage to set the output current.

$$<1>、I_{out} = \frac{17.7}{R_{ext}} * I_{gain} \quad @ I_{gain} < 67\% ;$$

$$V_{rext} = \frac{(I_{gain} - 67\%) * 1.19}{133\%} + 0.59$$

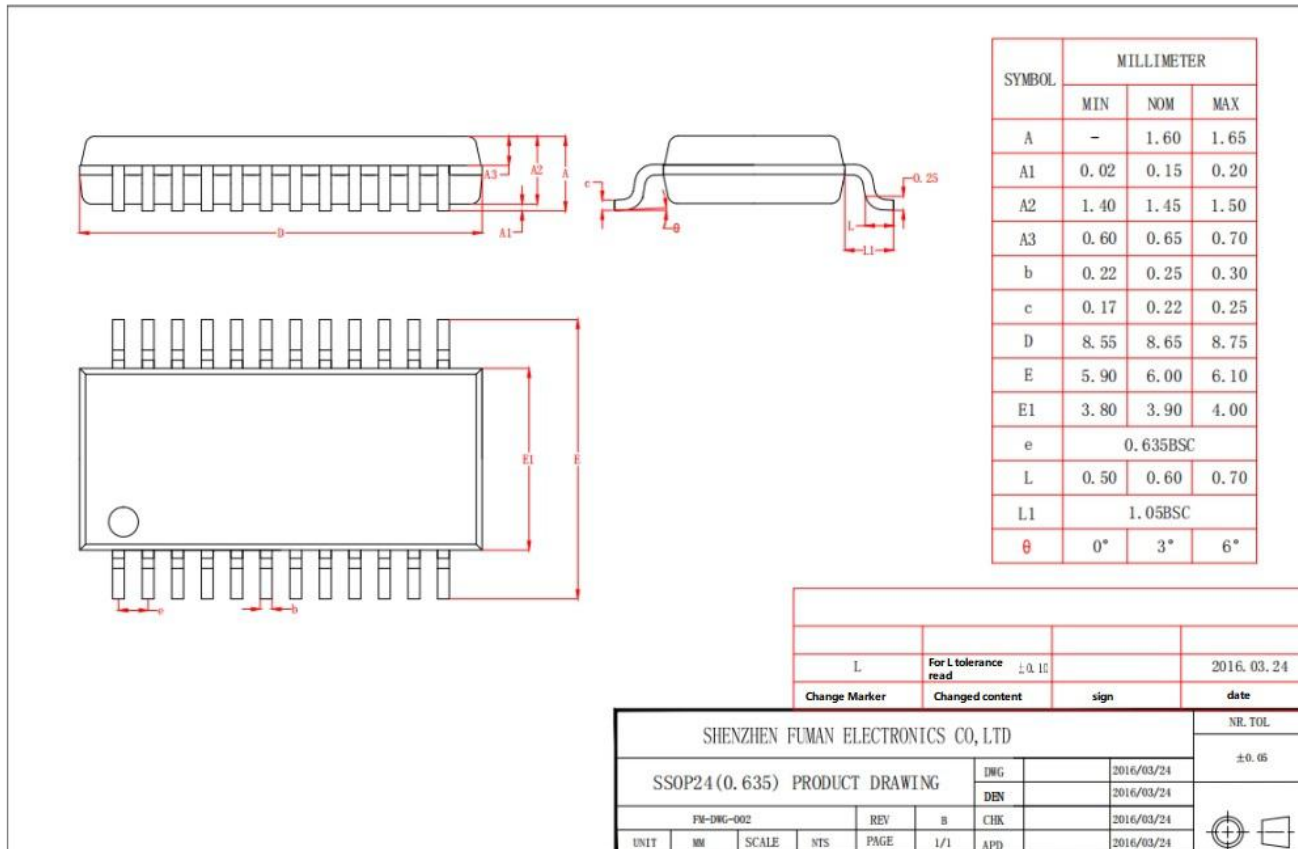
$$<2>、I_{out} = \frac{V_{rext}}{R_{ext}} * 20 \quad @ I_{gain} \geq 67\%$$

Vreft represents the voltage value of the REXT port, which can be configured by the internal register of the chip (the recommended voltage of VREXT is not less than 0.5V). Rext refers to the resistance value externally connected to the REXT terminal, and Igain represents the

current gain. The range is 22%-200%.

Package Information

SSOP24 (0.635)



QFN-24-4×4 (0.5mm)

